

# Omar AbdelKader

Lille, France



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## About Me

I am an AI researcher with extensive experience in artificial intelligence and software engineering. I am currently pursuing a Ph.D. in Computer Science at the University of Lille.

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## Responsibilities

### Leadership and Founding Roles

- Founder and Maintainer of the organization **pharo-llm**  on github and **pharo-llm**  on hugging face.
- President of **Synapse-NeuroTech-Lille** , a student association focused on AI and neuroscience.
- Project Leader and Maintainer for multiples projects **chatpharo** , **pharo-infer**  and many **more**.

### Community and Organizational Service

- Member of the organizing Committee, **GDR-SciLog'26** .
- Member of the organizing Committee, **ESUG'26** , **ESUG'25** , **ESUG'24** .
- Member in the organization **pharo-ai** .
- Member in **AI and Ecological Transition Club**  at **Ministry of Ecological Transition** .
- Contributor in **Café IA**  at **National Digital Council** .

### Reviewing, Mentoring, and Academic Supervision

- Sub reviewer at **BENEVOL'25** .

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## Awards and Honors

### Awards:

- Best Paper Award (3rd place) — IWST 2025, Gdańsk, Poland  
“Package-Aware Approach for Repository-Level Code Completion in Pharo” 

### Honors:

- Alumni Testimonial — IDMC 2024, University of Lorraine, France  
“Alumni testimonial in NLP & AI in University of Lorraine” 

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## Publications

### Journals

- AbdelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2026). Evaluating Package-Level Scoping Strategies for Repository-Level Code Completion in Pharo. *Journal of Computer Languages (COLA)*. DOI.

### Conferences

- AbdelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2025). Package-Aware Approach for Repository-Level Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.
- Elkolei, M., AbdelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.

- Kier, K., Giagnorio, A., Abdelkader, O., Zaitsev, O., Peharz, R., Robbes, R., Bavota, G., & Ducasse, S. (2026). Teaching LLMs a Low-Resource Language: Enhancing Code Completion in Pharo. *International Conference on Software Maintenance and Evolution (ICSME)*. DOI.

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## Experience

### PhD Student — INRIA, [EVREF], & Labri [Progress] (Villeneuve-d'Ascq, France) Oct. 2024 – Present

- Improving code completion and code generation using LLMs, specifically targeting the Pharo programming language, which has limited training data.
- Developing techniques for code completion, type inference, and deployment in Pharo's IDE, with a focus on runtime performance.

### AI Engineer — INERIS, [CIVS] (Verneuil-en-Halatte, France) Sept. 2023 – Sept. 2024

- Developed “INERIS-IA,” a tool to classify textual documents according to INERIS's strategic goals using ML and NLP techniques.
- Created Boolean queries for document retrieval and improved corpus quality through document similarity and keyword extraction.

### Intern — LIPN, [RCLN] (Villetaneuse, France) June 2023 – Aug. 2023

- Compared sampling techniques for probabilistic planning, particularly for generating literary narratives.
- Evaluated methods including the Score Function Estimator (SFE) and advanced techniques such as Gumbel-Softmax, assessing their effectiveness in producing coherent and creative stories.

### Software Developer — BEON-IT (Beirut, Lebanon) May 2022 – June 2022

- Java · Design Patterns · .NET Framework · Threading · Microsoft SQL Server

### Full-Stack Developer — SSCC-IT (Andket, Lebanon) Nov. 2020 – Apr. 2021

- PHP & SQL · Design Patterns · Microsoft SQL Server

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## Education

### University of Lille — Villeneuve-d'Ascq, France

- *Ph.D. in Computer Science* (Oct. 2024 – Present)

### University of Lorraine — Nancy, France

- *M.Sc. in Natural Language Processing* (Sept. 2022 – Sept. 2024)

### Lebanese University — Beirut, Lebanon

- *B.Sc. in Data Science* (Sept. 2019 – July 2022)

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## Software

### ChatPharo

- Live conversational interface between Pharo developers and large language models.
- Enables interactive discussions with LLMs directly from the Pharo environment.
- Designed an open and extensible architecture to experiment with multiple LLM backends.

### INERIS-IA

- AI platform developed at INERIS for document classification and knowledge management.
- Built a Flask-based web interface integrating multiple ML and NLP models.
- Trained document classification models aligned with COP 2027 objectives and INERIS strategic themes.
- Implemented additional features including document similarity, keyword extraction, and corpus quality improvement.

### Family Dynamics Analysis

- This project analyzes family dynamics in France using Formal Concept Analysis (FCA).

- Using data from the French national census, the study identifies patterns in family structures, single-parent households, childbearing decisions, and naturalization through marriage across five regional zones.
- The project explores relationships between demographic variables such as marital status, household composition, and nationality, offering insights into cultural and social trends shaping French households.

### Real-Fake Face Detection

- Demonstrates the process of building and training a neural network for classification tasks.
- Walks through data preprocessing, model architecture design, training, evaluation, and visualization of results to assess performance and prediction accuracy.

### EESMACF

- Evaluates two NLP models—SBERT (Sentence-BERT) and MiniLM—for classifying analogies using the FrameNet dataset.
- The objective is to identify valid and invalid analogies by leveraging SBERT embeddings and a fine-tuned MiniLM classifier.
- MiniLM achieved 99% accuracy in distinguishing valid from invalid analogies, outperforming SBERT (~55% accuracy).

### DeGatto

- Sentiment analysis framework for e-commerce women's apparel reviews.
- Uses a Kaggle dataset (23,000+ sentences) with aspect-level annotations for material, size, design, and comfort.
- Evaluated multiple NLP/DL/ML models (BiLSTM, SVM, Logistic Regression, Multinomial Naive Bayes). BiLSTM performed best for sentence-level analysis, and LinearSVC performed well for aspect-level analysis.
- Includes a visualization tool built with ReactJS and NodeJS to display results as bar or pie charts.

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## Teaching

- Teaching assistant at IUT: *Introduction to Software Development* 41,5 h Labs 
- Teaching assistant at IUT : *Software Maintenance* 12h Tutorials 
- Teaching assistant at Polytech: *DataBase* 16h Labs 
- Teaching assistant at Polytech: *Internet* 12h Labs 

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## Supervision

- Mehdi Elkolei, Bachelor Internship, 4 month, April 2026 - July 2026
- Srajal Dwivedi, Google Summer of Code, 3 month, May 2025 - July 2025

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## Public Talks

- **French Brain Awareness Week 2026:** AI and the brain: two ways of learning, what does that change for our daily lives? 
- **ESUG 2025:** ChatPharo: An Open Architecture for Understanding How to Talk Live to LLMs  

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## Technologies & Skills

- **Programming Languages:** Python, Pharo, Java, JavaScript
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Keras, Scikit-learn
- **Natural Language Processing (NLP):** spaCy, NLTK
- **Big Data & Distributed Systems:** Apache Spark, Hadoop
- **Data Analysis & Visualization:** Pandas, Matplotlib, Seaborn
- **Scientific Computing:** NumPy, SciPy
- **Web Development:** Flask, Django, Laravel, ReactJS
- **Databases & Cloud:** MongoDB, MySQL, AWS
- **Operating Systems:** macOS, UNIX, Windows
- **Research Areas:** Artificial Intelligence, Software Engineering, AI4SE, SE4AI

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## Volunteering

### President of Synapse-NeuroTech-Lille-ULille - (2024–Present)

- Led the executive board and coordinated association activities and projects.
- Managed the annual budget, including financial planning, expense tracking, and funding allocation.

- Represented the association in meetings with university administration, partners, and sponsors.
- Oversaw event planning, workshops, and conferences related to neurotechnology and artificial intelligence.
- Defined strategic goals and ensured alignment with the association's mission.

#### **Member in Club IA et Transition écologique at Ministère de la Transition Écologique — (2023–Present)**

- Contributed to ongoing AI-driven projects supporting ecological transition initiatives within the French government.
- Assisted in the development and evaluation of AI use cases for public policy and environmental impact.
- Collaborated with interdisciplinary teams, including policymakers, researchers, and technical experts.
- Participated in strategic discussions on responsible and sustainable AI deployment in the public sector.

#### **Contributor in Café IA at Conseil National du Numérique (CNNum) — (2023–Present)**

- Contributed to national discussions on artificial intelligence, digital policy, and societal impact.
- Assisted in the preparation and facilitation of Café IA sessions bringing together experts, researchers, and public stakeholders.
- Participated in the analysis of AI trends, use cases, and regulatory challenges.
- Supported knowledge-sharing initiatives on responsible and ethical AI.

#### **Volunteer First Aider in Croix-Rouge Française — (2022–2023)**

- Provided first aid and emergency assistance during public events and operations.
- Responded to medical emergencies following established safety and emergency protocols.
- Collaborated with multidisciplinary teams under high-pressure situations.
- Maintained readiness through regular training in first aid and emergency response.

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### **Hobbies**

- **Water Activities:** Swimming · Diving
- **Outdoor and Adventure Sports:** Hiking · Skiing · Skydiving
- **Creative and Artistic Activities:** Photography
- **Precision Sports:** Boxing · Bowling · Golf

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### **Languages**

- **Arabic:** Native
- **French:** Native
- **English:** B2

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Visit my website: [omarabedelkader.com](https://omarabedelkader.com)

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